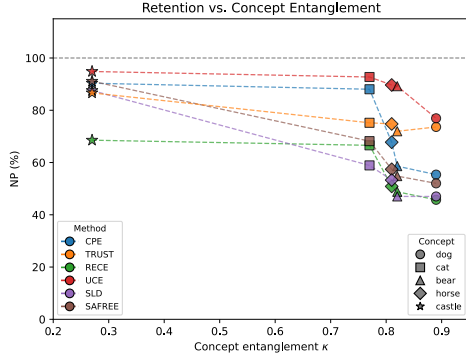


Table 12: **Additional baseline comparisons.** This table extends the averaged comparison in Table 11 with five newly evaluated baselines: UCE, RECE, CPE, TRUST, SLD, and SAFREE. Methods are grouped by intervention type: closed-form, fine-tuning, or inference-time. Cells report mean $\pm$ standard deviation across the evaluated target concepts, using the same erasure and retention metrics as the main table. The *Original* row denotes the unedited-model reference for this comparison.

Method	Type	Erasure		Retention			DamageGap $\downarrow$
		UA $\uparrow$	IRR $\downarrow$	IRA $\uparrow$	NP $\uparrow$	CP $\uparrow$	
uCE	Closed-form	80.6 $\pm$ 18.4	10.5 $\pm$ 23.4	67.1 $\pm$ 12.4	88.6 $\pm$ 25.1	99.3 $\pm$ 3.1	10.7 $\pm$ 25.4
RECE	Closed-form	99.4 $\pm$ 0.9	3.0 $\pm$ 4.0	43.3 $\pm$ 12.1	56.1 $\pm$ 10.6	93.5 $\pm$ 2.0	37.4 $\pm$ 10.2
CPE	FT	99.6 $\pm$ 0.9	4.7 $\pm$ 9.4	53.4 $\pm$ 12.4	71.9 $\pm$ 18.1	99.6 $\pm$ 2.6	18.7 $\pm$ 19.3
TRUST	FT	94.2 $\pm$ 2.2	6.5 $\pm$ 3.5	53.2 $\pm$ 13.7	76.4 $\pm$ 9.1	98.0 $\pm$ 5.0	11.6 $\pm$ 12.2
SLD	Inf.	99.6 $\pm$ 0.9	0.5 $\pm$ 0.9	35.0 $\pm$ 8.1	58.8 $\pm$ 16.8	89.1 $\pm$ 4.1	30.3 $\pm$ 16.3
SAFREE	Inf.	91.4 $\pm$ 11.4	8.4 $\pm$ 10.2	39.9 $\pm$ 8.6	64.7 $\pm$ 16.0	85.7 $\pm$ 12.2	21.0 $\pm$ 19.4



Concept	$\hat{\kappa}$	NP $\uparrow$	IRA-ratio $\uparrow$
castle	0.27	86.47	0.79
cat	0.77	74.88	0.78
horse	0.81	62.54	0.74
bear	0.82	59.71	0.70
dog	0.89	56.58	0.65
Spearman ρ		-1.0	-1.0

Figure 8: **Neighbor preservation scales inversely with $\hat{\kappa}$.** *Left:* NP for each (method, concept) pair. *Right:* New baseline’s NP and in-domain retention ratio across the five concepts (Spearman $\rho = -1.0$ for both).

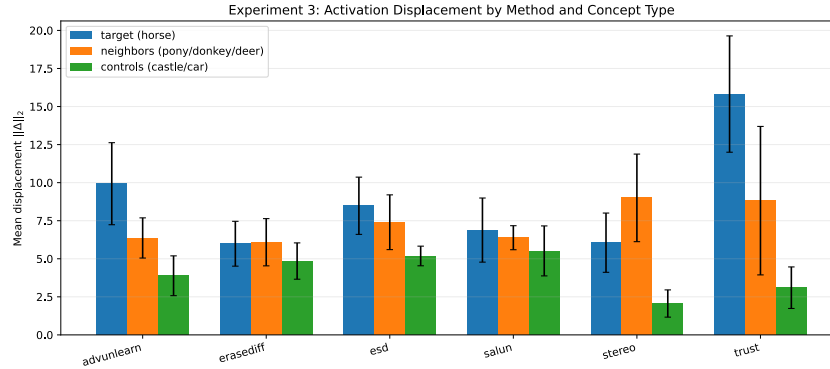


Figure 9: Mean activation displacement $\|\Phi_{\hat{\theta}}(p) - \Phi_{\theta}(p)\|_2$ for six unlearning methods erasing horse. Target prompts (blue) confirm Assumption 4.5; erasure requires measurable displacement. Neighbor prompts (orange) are displaced more than control prompts (green) for all methods, consistent with the propagation mechanism in Theorem 4.6.